

Ferrus Film-Coated Tablet 200mg

DESCRIPTION: A brown 7mm round, biconvex, plain on both sides, film-coated tablet

COMPOSITION: Each tablet contains of 200mg of Ferrous Fumarate.

PHARMACODYNAMICS: Iron is an essential component in the physiological formation of haemoglobin, adequate amounts of which are necessary for effective erythropoiesis and the resultant oxygen transport capacity of the blood. A similar function is provided by iron in Myoglobin production. Iron also serves as a co-factor of several essential enzymes. Orally taken iron, in the form of ferrous, passes through the mucosal cells as is bound with a protein called transferrin. In this form, iron is transported in the body to bone marrow for red blood cell production. Hence, Ferrous fumarate is an iron preparation that is used in the prevention and treatment of iron deficiency. A lack of iron affects the development of the red blood cells and causes reduction in the number of red blood cells found in the body (iron deficiency anaemia).

PHARMACOKINETICS:

Absorption

Iron is irregularly and incompletely absorbed from the gastrointestinal tract, the main sites of absorption being the duodenum and jejunum. Absorption is aided by acid secretion of the stomach and by some dietary acids (such as ascorbic acid) and is more readily affected when the iron is in the ferrous state or is part of the haem complex (haem-iron). Absorption is also increased in the conditions of iron deficiency or in the fasting state but is decreased if the body stores are overloaded. Normally only about 5 to 15% of the iron ingested in food is absorbed.

Distribution

Most absorbed iron is bound to transferrin and transported to the bone marrow where it is incorporated into haemoglobin; the remainder is contained within the storage forms, ferritin or hemosiderin, or as myoglobin, with smaller amounts occurring in haemo-containing enzymes or in plasma bound to transferrin.

Protein Binding

Very high (90% or more).

Haemoglobin – High

Myoglobin, enzymes and transferrin – Low

Ferritin and hemosiderin - Low

Elimination

No physiological system of elimination exists for iron, and it can accumulate in the body to toxic amounts; however, small amounts are lost daily in the shedding of skin, hair, and nails; and in feces, perspiration, breast milk (0.5 to 1 mg per day), menstrual blood, and urine.

Average daily loss of iron for healthy adults is:

Male	: 1 mg per day
Postmenopausal females	: 1.5 mg to 2 mg per day
Healthy premenopausal adult female	: 1.5 mg to 2 mg per day

INDICATION:

Iron Deficiency Anaemia (Prophylaxis and Treatment)

Iron tablets are indicated in the prevention and treatment of iron deficiency anaemia, which may result from inadequate diet, malabsorption, pregnancy, rapid growth during childhood, and/or blood loss

RECOMMENDED DOSAGE

	Dosage	Administration
Prophylactic	200 mg daily	Oral
Therapeutic	200 mg three or four times a day, the dosage being adjusted gradually as needed and tolerated	Oral

- This medicine is best taken on empty stomach, 1 hour before or 2 hours after meals for better absorption; but may be taken with food to lessen possibility of stomach upset.
- Avoid regular use of large amounts of iron supplements several times daily for more than six months unless approved by physician.
- Any signs of overdose must be immediately reported to physician.

ROUTE OF ADMINISTRATION: Oral

CONTRAINDICATIONS:

Except under special circumstances, this medication should not be used when the following medical problems exist:

- Hemochromatosis
- Hemosiderosis
- Other anaemic conditions, unless accompanied by iron deficiency (some conditions such as haemolytic anaemia or thalassemia, may cause excess storage of iron)
- Porphyria cutanea tarda

WARNINGS & PRECAUTIONS:

Iron compounds should not be given to patients receiving repeated blood transfusions or to patients with anaemias not produced by iron deficiency unless iron deficiency is also present. Oral and parenteral iron therapy should not be used together. Care should be taken in patients with iron-storage or iron-absorption disease such as haemochromatosis, haemoglobinopathies, or existing gastrointestinal disease such as inflammatory bowel disease, intestinal strictures and diverticulae.

Geriatrics

Problems in geriatrics have not been documented with intake of normal daily recommended amounts. Some geriatrics patients may require a larger than usual daily ingestion of bioavailable iron to correct an iron deficiency, because of their ability to absorb iron has been diminished by reduced gastric secretions and achlorhydria.

INTERACTIONS WITH OTHER MEDICAMENTS:

Acetohydroxamic acid

Iron and possibly other heavy metals, when taken normally, are chelated by acetohydroxamic acid; this may result in reduced intestinal absorption of both acetohydroxamic acid and oral iron.

Alcohol

Concurrent use with ferric iron for a prolonged period may result in toxicity since absorption and hepatic storage of iron are increased, especially if alcohol usage is high.

Antacids; Calcium Supplements; Coffee; Eggs; Food or Medications containing Bicarbonates, Carbonates, Oxalates, or Phosphates; Milk or Milk Products; Tea containing Tannic Acid; Whole-grain Breads and Cereals; Dietary Fibre

Concurrent use with iron may decrease iron absorption because of the formation of less soluble or insoluble complexes.

Cimetidine

The decrease in gastric acid caused by cimetidine may decrease the absorption of nonheme iron; concurrent use with iron supplements is not recommended.

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Deferoxamine and Possibly Other Chelating Agents

Deferoxamine chelates iron and is used in the treatment of iron overdose and other iron overload conditions; iron may be necessary in patients receiving other chelating agents; however, it should be given at least 2 hours after chelating agents.

Dimercaprol

Concurrent administration of medicinal iron with dimercaprol results in the formation of a toxic complex; if iron deficiency is present, its treatment should be postponed until therapy with dimercaprol has been discontinued for at least 24 hours; severe iron deficiency anaemia occurring during dimercaprol therapy should be managed with blood transfusion.

Etidronate

Concurrent use may prevent absorption of oral etidronate; patients should be advised to avoid using iron supplements within 2 hours of etidronate.

Fluoroquinolones

Iron may reduce absorption of fluoroquinolones by chelation, resulting in lower serum and urine concentrations of fluoroquinolones.

Penicillamine or Trientine

Concurrent use with iron may decrease the therapeutic effects of these medications; if necessary, iron may be administered in short courses, but a period of 2 hours should elapse between administration of penicillamine or trientine and iron.

Oral Tetracyclines

Concurrent use with iron reduces absorbability and resultant therapeutic effects of oral tetracyclines; patients should be advised to take iron supplements 2 hours after tetracycline.

Oral Zinc Supplements

Large doses of iron supplements have been found to inhibit the intestinal absorption of zinc; this may be a problem in individuals taking commercial multivitamin-mineral preparations or infant formulas that have a high iron-to-zinc ratio; zinc supplements should be taken at least 2 hours after iron supplements.

USE IN PREGNANCY & LACTATION:

Use in Pregnancy: In the first trimester of pregnancy, adequate iron intake is usually obtained from a proper diet; however, in the second and third trimesters, when iron deficiency is more prevalent because of greatly increased requirements, iron supplements may be recommended.

Use in Lactation: Problems in humans have not been documented with intake of normal daily recommended amounts.

SIDE EFFECTS: The astringent action of oral iron preparations sometimes produces gastrointestinal irritation and abdominal pain with nausea and vomiting. These irritant adverse effects are usually related to the amount of elemental iron taken rather than the type of preparation. Other gastrointestinal effects may include either diarrhoea or constipation. The stools commonly become dark green or black when iron preparations are taken orally. This is caused by the presence of unabsorbed iron and is harmless.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Symptoms

Early symptoms of acute iron toxicity

Diarrhoea, sometimes containing blood; fever; severe nausea; sharp stomach pain or cramping; severe vomiting, sometimes containing blood.

Late symptoms of acute iron toxicity

Bluish-coloured lips, fingernails, palms of hands; drowsiness; pale, clammy skin; seizures; unusual tiredness or weakness; weak and fast heartbeat.

Treatment of Acute Overdose

Overdose of ingested iron can be fatal, especially in small children. Immediate treatment is essential. Serious poisoning may result in small children from ingestion of 200 mg of elemental iron. After one hour, excessive systemic absorption of iron and possible erosion of stomach and intestinal tissues complicate evacuative and supportive procedures. Transport of patient to emergency room should not be delayed. If syrup of ipecac has been administered, emesis may require up to 30 minutes or even a repeat dose. However, patient transport must not wait for emetic effect.

Overdose symptoms may be delayed (10 to 60 minutes or longer) because of many intervening factors such as iron salt taken, amount of food in stomach, and size of dose. To decrease absorption includes inducing emesis with syrup of ipecac or lavaging with sodium bicarbonate if patient is comatose or having convulsions may be used, depending on patient's condition. If intact radiopaque tablets are visualised on x-ray, repeating lavage may be necessary.

Laboratory studies on heparinized blood should include serum iron, haemoglobin, haematocrit, electrolytes, blood gases and blood glucose, total iron-binding capacity (TIBC), complete blood count, blood type, and cross-match. Serum iron determinations should be repeated. Serum drawn early (within 2 hours of ingestion) may have artificially high concentrations of iron. Peak serum concentrations are reached about 6 hours after ingestion. However, achievement of peak serum concentrations is delayed if the iron was in extended-release form, if the patient had a significant amount of food in his/her stomach, and if the dose of iron was large.

Specific treatment includes maintaining of fluid and electrolyte balance. Acidosis may be corrected with intravenous sodium bicarbonate. Antidote: Deferoxamine, administered slowly, intravenously or intramuscularly, is used in more severe iron toxicity, when symptoms are other than minimal vomiting or diarrhoea. Dialysis is of no value in removing serum iron alone, but may be used to increase excretion of the iron-deferoxamine complex, and is indicated in the presence of anuria or oliguria. Exchange transfusion may be successful. Patient must be observed for a minimum of 24 hours after becoming asymptomatic. Delayed effects may include shock and severe gastrointestinal bleeding (24 to 48 hours), and gastrointestinal obstruction (weeks to months). Residual damage may be ruled out with liver and upper gastrointestinal studies.

STORAGE CONDITIONS: Keep in a dry place. Store below 30 °C. Protect from light. Keep out of reach from children. *Jauhkan daripada kanak-kanak.*

SHELF LIFE:

Please refer to outer package.

PACK SIZE:

Blister pack of 30 tablets.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

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Taman Klang Jaya, 41200 Klang, Selangor Darul Ehsan, Malaysia.

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